

Proposed Extensions

1. Multimodal GraphRAG

Problem: Current GraphRAG primarily retrieves information from textual content and cannot effectively utilize information present in tables, diagrams, and images.

Extension:

- Extract information from:
 - Text
 - Tables
 - Diagrams/Images
- Convert extracted information into graph nodes and relationships.
- Enable retrieval across multiple modalities.

Example:

Motor Temp = 95°C (table)

Motor → Cooling → Shutdown (diagram)

The system combines information from both sources to answer:

"The motor stopped because its temperature exceeded the threshold, triggering the shutdown mechanism."

2. Critic Agent (Self-Correcting GraphRAG)

Problem: LLM-generated answers may contain hallucinations, unsupported claims, or incomplete information.

Extension:

- Introduce a Critic Agent after answer generation.
- Verify:
 - Is the answer supported by retrieved evidence?
 - Is any important information missing?
 - Is there any contradiction with the source documents?
- If validation fails, trigger re-retrieval and answer regeneration.

Workflow:

Question



GraphRAG



Answer



Critic Agent



PASS / FAIL

If **FAIL**, retrieve additional evidence and regenerate the response.

3. Cascaded Dynamic GraphRAG

Description:

A query-routing mechanism dynamically selects between Naive RAG and GraphRAG based on query complexity. Simple factual questions are answered using lightweight vector retrieval, while multi-hop reasoning questions are routed to GraphRAG. This reduces latency and computational cost while maintaining answer quality.

Example:

What is motor threshold?

→ Naive RAG

Why did motor overheating cause shutdown?

→ GraphRAG

4. Explainable GraphRAG

Problem: Users cannot easily understand how the system arrived at a particular answer.

Extension:

- Provide an explainable reasoning trace for every answer.
- Display:
 - Source document
 - Page number
 - Retrieved chunk(s)
 - Graph traversal path
 - Supporting evidence

Example:

Battery_Manual.pdf

Page 24

↓

Chunk 145

↓

Overheating

↓

Thermal Runaway

↓

Safety

This enables users to verify the answer source and understand the reasoning process.

5. Knowledge Graph Visualization Dashboard

Problem: The knowledge graph and retrieval process remain hidden from users.

Extension:

- Develop an interactive graph visualization dashboard.

- Visualize:
 - Documents
 - Chapters
 - Chunks
 - Keywords
 - Tables
 - Images
- Highlight nodes and edges involved in answering a query.

Example:

Robot Arm



Motor Fault



Overheating



Shutdown

The retrieval path is highlighted in real time, allowing users to observe how the answer was generated.
